

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester II Examination, 2023-24

Paper No. MTB-202: Statics & Dynamics

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 074904

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each.)

- (a) Obtain a single force or couple equivalent to four forces with components $(2, 3)$, $(3, -1)$, $(-4, -4)$, $(-1, 2)$ acting at $(0, 1)$, $(1, -1)$, $(2, 3)$, $(-1, -2)$ respectively.
- (b) Find the work done by the thrust at the ends of a rod when displacement alters its length.
- (c) Prove that a system of coplanar forces is in equilibrium if the algebraic sum of moments about each of three non-collinear points is zero.
- (d) A particle of mass m moves along a path with linear velocity v at distance r from the origin. Obtain the angular velocity ω .
- (e) Which of the following are NOT properties of SHM: (i) acceleration always directed to a fixed line; (ii) constant distance from a fixed point; (iii) constant linear velocity; (iv) motion always in a straight line.
- (f) If radial and transverse velocity components are λr and μr^2 (λ, μ constants), find the polar equation of the path.
- (g) State Kepler's third law in mathematical form and prove it.
- (h) A particle describes a cardioid $ap^2 = r^3$ under a central force. Find the law of force.
- (i) Is areal velocity of a particle under central force always proportional to angular momentum per unit mass? Give reasons.
- (j) A particle of mass 3 units moves up the inner surface of a smooth vertical circular path of radius 5 with initial velocity 15. Find the reaction at angle 60° ($g = 9.8$).
- (k) A frame has linear motion with acceleration. Is it a Newtonian frame of reference? Give reasons.

Question 2.

- (a) Show that a system of coplanar forces can be reduced to a single force through a given point and a couple. [6]

- (b) A system of forces in the plane of triangle ABC is equivalent to a single force at A along the internal bisector of $\angle BAC$ and a couple G_1 . If moments about B and C are G_2 and G_3 , show $(b + c)G_1 = bG_2 + cG_3$, where $b = CA$, $c = AB$. [6]

Question 3.

- (a) State and prove the principle of virtual work for a system of coplanar forces. [6]
- (b) A heavy elastic string of natural length $2\pi a$ is placed round a smooth cone of semi-vertical angle θ , axis vertical. Weight W , modulus λ . Prove it is in equilibrium when it forms a circle of radius $a\left(1 + \frac{W \tan \theta}{2\pi\lambda}\right)$. [6]

Question 4.

- (a) Derive the tangential and normal components of acceleration of a particle moving in a plane curve. Also write the radial and transverse components. [6]
- (b) A point moves in a curve so that tangential and normal accelerations are equal and the tangent rotates with constant angular velocity. Show the intrinsic equation is $s = Ae^\theta + B$. [6]

Question 5.

- (a) Show that in SHM the average speed and average acceleration are obtained by multiplying their maximum values by $\frac{2}{\pi}$. [6]
- (b) Discuss the motion of a particle in a frame rotating with constant angular velocity about a fixed frame and obtain the fictitious forces. [6]

Question 6.

- (a) If the central orbit is $r^3 = a^3 \cos 3\theta$ under force toward the pole, find the law of force and velocity at any position. [6]
- (b) A particle under central acceleration $\mu/(\text{distance})^2$ is projected with velocity V at distance R . Show its path is a rectangular hyperbola if angle of projection is $\sin^{-1}\left\{\mu/(VR\sqrt{V^2 - 2\mu/R})\right\}$. [6]

Question 7.

- (a) A particle moves on a smooth vertical circular wire of radius a , projected from the lowest point with velocity $\sqrt{4ga}$. Show the reaction is zero after time $\sqrt{a/g} \log(\sqrt{5} + \sqrt{6})$. [6]

- (b) Two particles drop from the cusp of a cycloid at interval t . Prove they meet after time $2\pi\sqrt{a/g} + t/2$. [6]